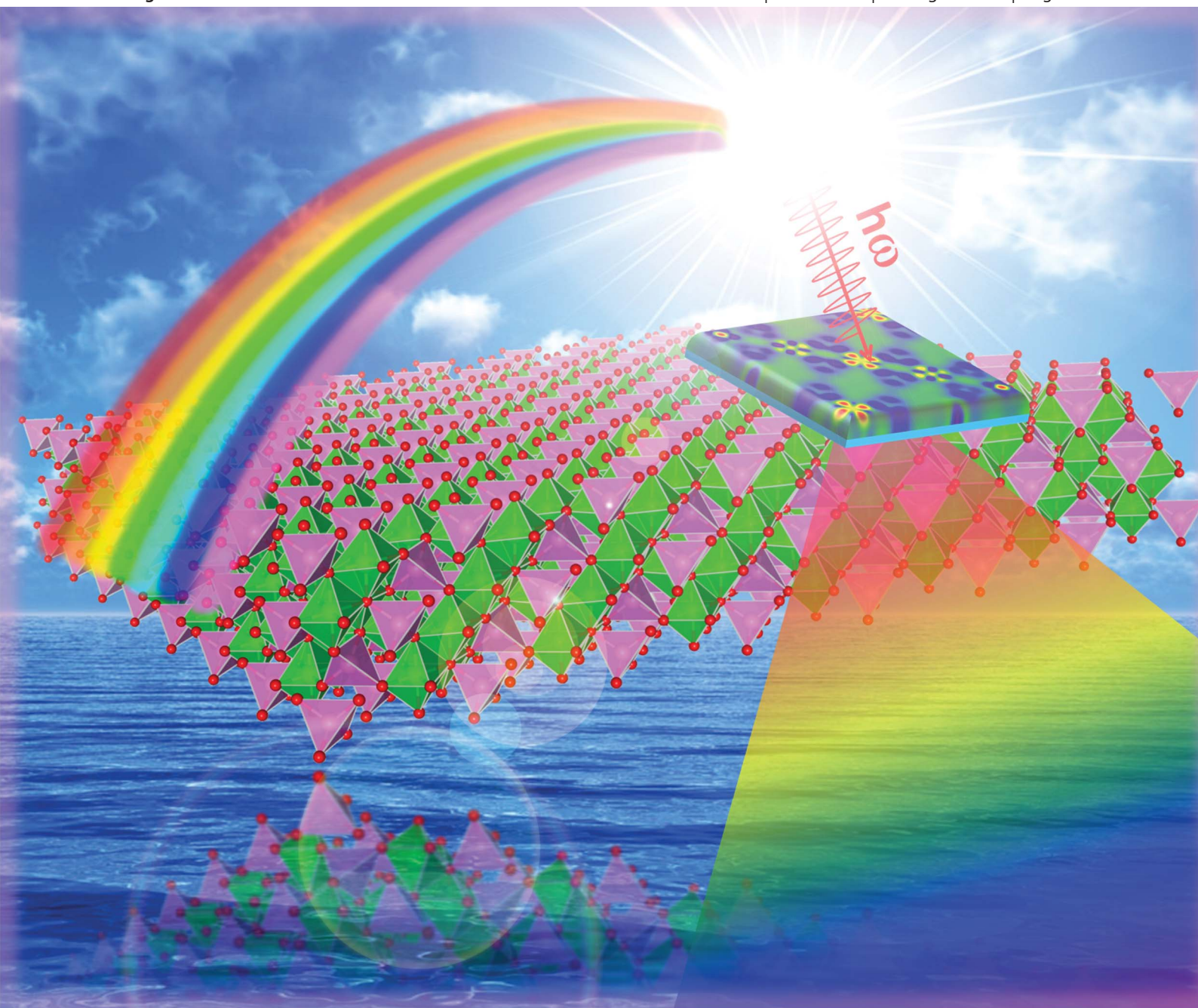


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L. Qiao, M. D. Biegalski *et al.*

Nature of the band gap and origin of the electro-/photo-activity of Co_3O_4



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L. Qiao,^{*a} H. Y. Xiao,^b H. M. Meyer,^c J. N. Sun,^d C. M. Rouleau,^a A. A. Poretzky,^a
D. B. Geohegan,^a I. N. Ivanov,^a M. Yoon,^a W. J. Weber^b and M. D. Biegalski^{*a}

Co_3O_4 exhibits intriguing physical, chemical and catalytic properties and has demonstrated great potential for next-generation renewable energy applications. These interesting properties and promising applications are underpinned by its electronic structure and optical properties, which are unfortunately poorly understood and the subject of considerable debate over many years. Here, we unveil a consistent electronic structural description of Co_3O_4 by synergetic infrared optical and *in situ* photoemission spectroscopy as well as standard density functional theory calculations. In contrast to previous assumptions, we demonstrate a much smaller fundamental band gap, which is directly related to its efficient electro-/photo-activity. The present results may help to advance the fundamental understanding and provide guidance for the use of oxide materials in photocatalysis and solar applications.

The ever-increasing needs of global energy exploration demand alternative energy conversion and storage systems that are clean, efficient, and renewable. Spinel Co_3O_4 has recently been shown to exhibit remarkable photo- and electro-chemical properties as well as an excellent stability during oxygen reduction and evolution reaction processes.^{1–6} These properties, combined with its cost effectiveness and wide abundance, have promoted this material as a promising candidate for fuel cell, lithium ion batteries, water splitting and artificial photosynthesis energy applications. However, despite its promise,^{7,8} the basic electronic structure of Co_3O_4 that is responsible for these intriguing materials properties is poorly understood both experimentally and theoretically. Particularly, the nature and

size of the fundamental band gap^{9–18} as well as the origin of the various optical excitations that cover the entire solar spectrum are still under serious debate.^{12,19–25} These discrepancies are major barriers that limit the potential of this material for important energy applications, and most likely result from the lack of a comprehensive understanding of its intrinsic optical properties and the absence of a reliable theoretical method for the description of its structure and properties.

In this work, we address previous issues with the electronic structure through the use of phase pure, single crystalline, epitaxial thin films of Co_3O_4 (see the ESI, Fig. S1[†]), coupled with an examination of the optical and electronic properties across a broad wavelength range extending deep into the NIR, to create an accurate model for the first-principle simulation of its structure and properties. For the first time, we clearly demonstrate that Co_3O_4 exhibits a fundamental band gap of ~ 0.74 eV, much smaller than the generally accepted values (*e.g.* from 1.5 to 2.5 eV (ref. 9–18)) and this gap originates from a direct dipole-forbidden d–d transition between tetrahedral-site Co^{2+} cations. Our results lead to a full description of the electronic structure of Co_3O_4 , and reveal that it is not a highly correlated material. These discoveries, coupled with its strong light absorption across the solar spectrum and p-type semiconducting nature, directly account for its excellent efficiency in various electro-/photo-related chemical processes and provide new insights for designing novel nano- and hetero-structures for energy applications. More importantly, we present a universal combined experimental and theoretical approach to unveil the details of the complicated electronic structures and optical properties of complex oxides, which can be extended to other multiband materials exhibiting complex optical absorptions.

High-quality relaxed epitaxial Co_3O_4 films were grown on $\text{Al}_2\text{O}_3(0001)$, $\text{MgO}(001)$ and $\text{SrTiO}_3(001)$ substrates by pulsed laser deposition from high-purity (99.9985%) stoichiometric Co_3O_4 targets. Films were deposited with a 248 nm excimer laser source with an energy density of 1.5 J cm^{-2} , at a substrate temperature of 600°C , and an oxygen partial pressure of 200 mTorr, and were cooled down to room temperature in

^aCenter for Nanophase Materials Sciences, Oak Ridge National Laboratory, Oak Ridge, TN 37831, USA. E-mail: qiao@ornl.gov; biegaliskim@ornl.gov

^bDepartment of Materials Science and Engineering, University of Tennessee, Knoxville, TN 37996, USA

^cMaterials Science and Technology Division, Oak Ridge National Laboratory, Oak Ridge, TN 37831, USA

^dJ. A. Woollam Co., Inc., 645 M Street, Lincoln, NE 68508, USA

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200 Torr pure oxygen. The structure and quality of the epitaxial films were determined by reflective high-energy electron diffraction, X-ray diffraction and atomic force microscopy. The electronic and optical properties of the films were characterized by X-ray photoemission, optical absorption, photoluminescence, photoluminescence excitation, and ellipsometry.

Density of states (DOS) and band structures were studied by density functional theory (DFT) calculations using generalized gradient approximation with a Perdew–Burke–Ernzerhof function based on the Vienna Ab initio simulation package.²⁶ A Hubbard U correction for the on-site Coulomb interaction was considered, with effective U values ranging from 0 to 6 eV for both Co ion species. All calculations included spin-polarization, in which an antiferromagnetic high-spin structure for the tetrahedral Co^{2+} ion and a nonmagnetic low-spin structure for the octahedral Co^{3+} ion were employed. An $8 \times 8 \times 8$ k -point sampling in reciprocal space and a plane-wave basis set with an energy cutoff of 500 eV were used and the total energy was minimized to ensure the force on each ion was converged to less than $0.01 \text{ eV } \text{\AA}^{-1}$.

The optical absorption coefficient (α) for a 17 nm epitaxial Co_3O_4 film was measured within a wide photon energy range between 0.45 and 6.2 eV to include the NIR optical response,²⁷ as shown in Fig. 1(a). Four distinct absorption bands were clearly revealed with threshold energies of 0.7, 1.6, 2.2, and 4.4 eV. The lowest energy transition band exhibited three peaks, as marked in Fig. 1(b). At a higher energy there was a steep increase in the absorption with a second pronounced band at 1.6 eV. This sharp peak exhibited a highly asymmetric line-shape, indicative of multiple absorption bands. For energies

above 2.2 eV, the two absorption bands were much stronger in intensity and broader than the first two, and clearly exhibited visible knee-like features at 3.7 and 4.6 eV. Based on our previous approach,²⁸ the overall absorption spectrum can be well fitted by a convolution of nine Gaussians accounting for the film absorption (peaks 1–9), one Gaussian for the substrate contribution (peak 10), and a broad low-energy feature due to the refractive index mismatches at the air–film and film–substrate interfaces (peak 11). The existence of this extrinsic contribution (peak 11) was further confirmed by ellipsometry measurements (see the ESI, Fig. S2†).

Although Co_3O_4 is highly absorbing, it is not a highly anisotropic optical material, because the absorption features for $\text{Co}_3\text{O}_4(111)$ films on $\text{Al}_2\text{O}_3(0001)$ are nearly identical with $\text{Co}_3\text{O}_4(001)$ films grown on $\text{MgO}(001)$ or $\text{SrTiO}_3(001)$ (see the ESI, Fig. S3†). Thus, in this work we focus on the films grown on $\text{Al}_2\text{O}_3(0001)$, although similar results were found for different orientations of Co_3O_4 grown on other substrates. Based on a line extrapolation of the lowest absorption peak in the Tauc plot of Fig. 1(b), the optical band gap of Co_3O_4 was determined to be 0.76 eV. A similar analysis of the stronger, next higher-energy transition would yield a band gap of 1.6 eV consistent with previous data.¹⁵ It is important to note that the lower band gap of 0.76 eV reported here required measurements extending into the NIR. This band gap was supported by infrared photoluminescence (PL) measurements of a ~ 100 nm thick epitaxial Co_3O_4 – $\text{SrTiO}_3(001)$ at 10 K [Fig. 1(c)]. Under excitation by a 404.5 nm laser, there was clearly a sharp asymmetric peak at 0.74 eV. This emission does not arise from the substrate, which showed a broad and higher energy peak [Fig. 1(c)]. A similar

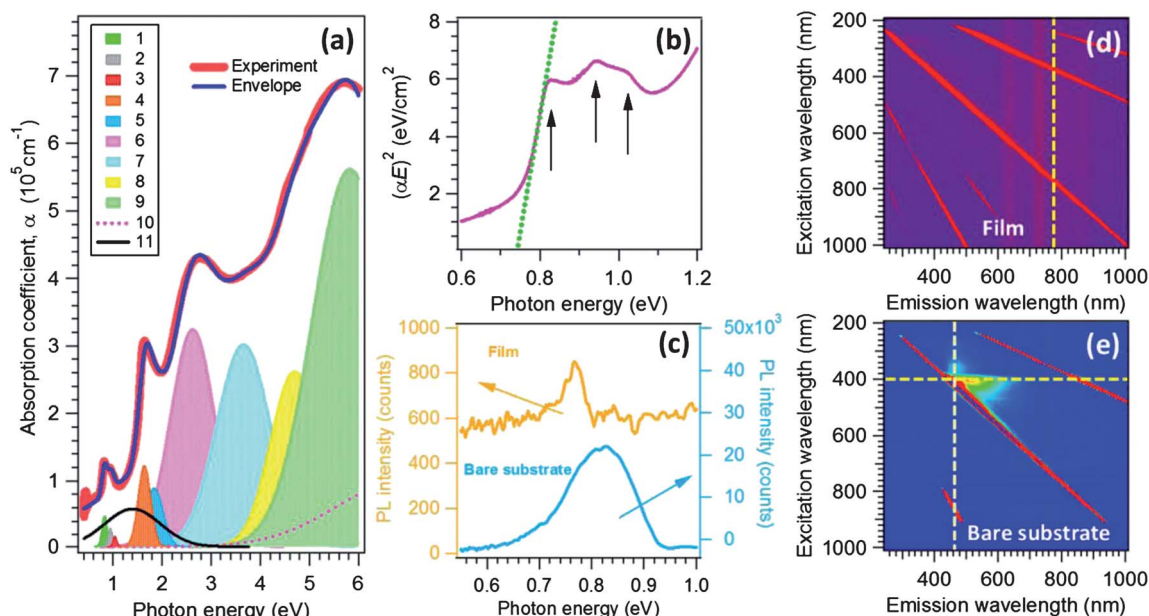


Fig. 1 Optical absorption and photoluminescence properties. (a) Experimental wide photon energy range optical absorption for a 17 nm epitaxial $\text{Co}_3\text{O}_4(111)$ film, along with Gaussian peak deconvolutions. The thick red line is the experiment data and the thin blue line is the peak envelope. The optical transition assignments are listed in Table 1. (b) Tauc plot of the same spectrum near the band gap region, where three additional peaks can be seen, as labelled by the black arrows. The band gap of ~ 0.74 eV was determined by a linear extrapolation of the absorption onset, as indicated by the dotted green line. (c) 10 K PL spectrum of Co_3O_4 – SrTiO_3 and the bare SrTiO_3 substrate under a 404.5 nm laser excitation. Photoluminescence excitation spectra mapping for (d) epitaxial Co_3O_4 – SrTiO_3 and (e) the bare SrTiO_3 substrate.

low-energy in-gap PL that can be attributed to defects or surface states in the SrTiO₃ substrate has also been reported.²⁹ Based on UV-vis photoluminescence excitation (PLE) spectral mapping [Fig. 1(d) and (e)], it can be seen that the Co₃O₄ film does not exhibit any detectable emission near 1.6 eV (~770 nm), where PL might originate if the band gap was in this region (note: there are two emissions from the lamp source at 730 and 620 nm). In addition, the sharp absorption edge and relatively small Stokes shift indicate that the band gap may correspond to a direct-type transition. Though the influences of defect states and substrate effects can not be completely ruled out, the PL emission at 0.76 eV for the Co₃O₄ is consistent with absorption, ellipsometry, and is supported in the subsequent arguments through XPS and the following sections, strongly indicating the fundamental band gap is below 1.6 eV.

To clarify the electronic structure of Co₃O₄, which contains heterovalent Co²⁺ and Co³⁺ cations, we first compared it to the structure of CoO to separate the electronic structure due to the different valance states of the cobalt ions. Fig. 2(a) and (b) display the Co 2p core-level (CL) and valence band (VB) X-ray photoelectron spectroscopy (XPS) spectra of Co₃O₄ and the reference epitaxial CoO, which revealed significant differences between the electronic structures. These differences were demonstrated by the much sharper CL and VB spectra for Co₃O₄. Specifically Co₃O₄ exhibited a distinctive feature, ②, at the top of the VB with a clear shoulder (feature ③) at a lower binding energy, while CoO exhibited much broader peaks due to multiple splitting from unpaired 3d electrons and p–d charge transfer satellite splitting.³⁰ To examine these differences in detail, Co₃O₄ was sputtered *in situ* using Ar⁺ in a high vacuum to reduce the surface of the sample to CoO; XPS analysis confirmed the similar broad features of the sputtered Co₃O₄ and reference CoO (see the green dotted and blue solid lines). A comparison of the Co₃O₄ VB spectra before and after sputtering in Fig. 2(b) revealed an obvious increase in the photoelectron yield near the valence band maximum (VBM), a significant

intensity decrease at 1 eV below the VBM, and an intensity increase at higher binding energies. Therefore, the features ① and ③ of Co₃O₄ can be ascribed to the Co²⁺ photoelectrons, while the sharp feature ② corresponds to Co³⁺. In addition, the features ④ and ⑤ are ascribed to O 2p and Co³⁺ satellites. These peak assignments partially agree with the work done by Chuang *et al.*³¹ and are quite consistent with the cluster calculation results by Jugnet and Duc.²⁵

With detailed measurements of the electronic structure, first-principle calculations could be employed to determine the electronic fine structures near the Fermi level. The dependency of the total and partial DOS on different effective Hubbard *U* correction values (*U*_{eff}), for both Co²⁺ and Co³⁺ were calculated and the results are shown as two-dimensional maps in Fig. 3(a)–(d) for the total, Co²⁺ 3d, Co³⁺ 3d, and O 2p orbitals, respectively. Generally, with increasing *U*_{eff} values for both Co ions, the energy gap between the VBM and the conduction band minimum (CBM) expanded quickly [Fig. 3(a)]. For example, plus *U* calculations with *U*_{eff}(Co²⁺) > 2 eV and *U*_{eff}(Co³⁺) > 1 eV led to a band gap larger than the experimental value of 0.76 eV (see the ESI, Fig. S4a†), and it grew to ~2.2 eV for higher *U*_{eff} values of *U*_{eff}(Co²⁺) = 4 eV and *U*_{eff}(Co³⁺) = 6 eV. For standard DFT, the band gap was smaller than the experimental value due to the well-known DFT underestimation. On the other hand, the increased *U*_{eff} values led to increasing discrepancies in lattice constants from the experiments, while the differences between DFT + *U* with small *U*_{eff} and standard DFT were small (see ESI, Fig. S4b†).

However, the Hubbard *U* correction with large *U*_{eff} values introduced remarkable distortions in the electronic structure. When *U*_{eff}(Co³⁺) exceeded 1 eV, there was a clear discontinuity at the VBM for the total DOS due to the downward bending of the edge corresponding to Co³⁺ 3d orbitals, and this gap was further increased with larger *U*_{eff}(Co²⁺) values [see the circles in Fig. 3(a) and (c)]. Similarly, when *U*_{eff}(Co²⁺) exceeded 1 eV, the DOS for the Co²⁺ 3d orbitals was pushed away from the VBM [see the triangles in Fig. 3(b)], which is inconsistent with the photoemission Co²⁺ VBM feature in Fig. 2(b). The calculated total and partial DOS with two extreme combinations of *U*_{eff} values can be directly compared with the experimental XPS VB spectra in Fig. 4(a) and (d), where the electronic structure distortion caused by the Hubbard *U* correction can be clearly visualized. Since first-principle methods calculate the ideal DOS distribution at each energy level without considering the actual lifetime broadening of the electron–hole pairs and instrumental broadening of the spectrometer, the calculated DOS was not expected to predict the exact lineshape of the VB. Nevertheless, the calculated energy positions were expected to be close to the experimental values.³² Note that standard DFT with *U*_{eff} = 0 eV for Co²⁺ and Co³⁺, resulted in a total DOS distribution quite consistent with the experimental VB features at each energy position, as shown in Fig. 4(d). However, in the case of *U*_{eff}(Co²⁺) = 4.4 eV and *U*_{eff}(Co³⁺) = 6.7 eV, which have been suggested to reproduce the 1.6 eV band gap,²⁴ the VB electronic structure was significantly distorted, as shown in Fig. 4(a). Other intermediate *U*_{eff} value combinations also induced different extents of these VB distortions, as indicated in Fig. 4(b) and (c).

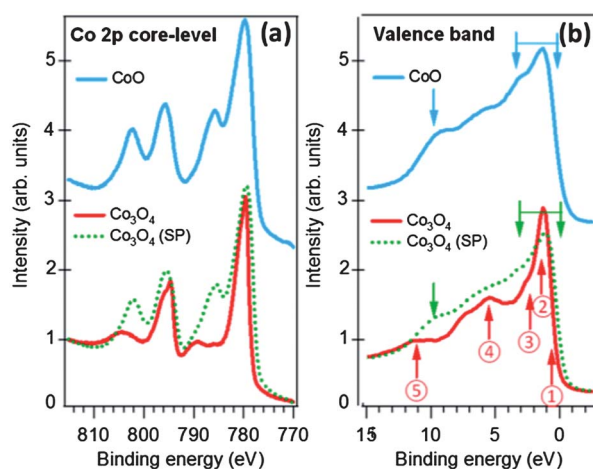


Fig. 2 *In situ* photoelectron spectroscopy. XPS Co 2p core-level (a) and valence band (b) spectra for the Co₃O₄(111) and CoO(111) films. The dotted lines are for the Co₃O₄(111) film after *in situ* Ar⁺ sputtering (SP).

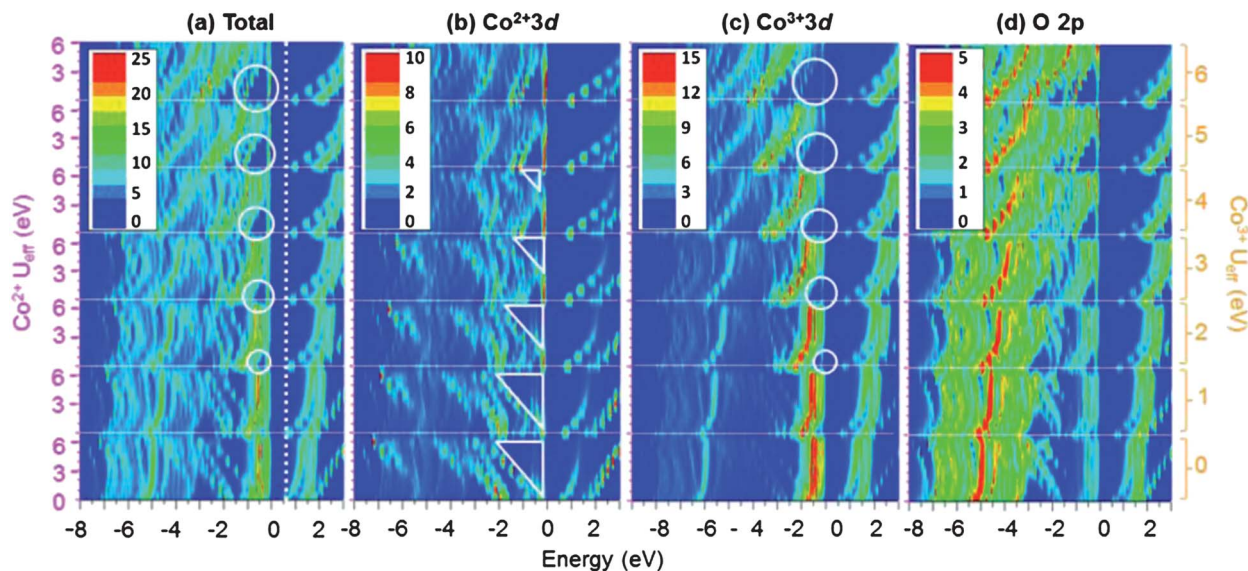


Fig. 3 DFT calculations under different applied U_{eff} values. Two-dimensional contour mapping of the calculated DOS as a function of different U_{eff} values for both Co^{2+} (left scale) and Co^{3+} (right scale) ions projected onto (a) the total, (b) Co^{2+} 3d, (c) Co^{3+} 3d, and (d) O 2p orbitals. The vertical dotted line in (a) indicates the experimental band gap of 0.76 eV. The intensity scales are labeled in sets.

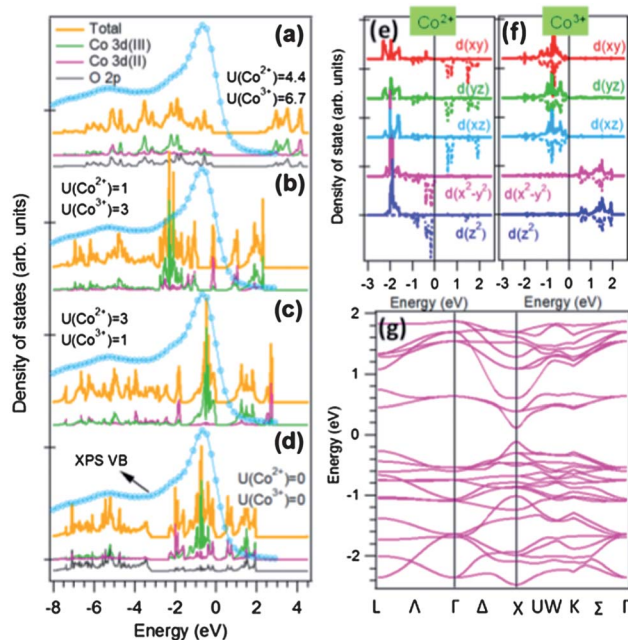


Fig. 4 Electronic fine structure near the Fermi level. Comparison of XPS VB spectra with the calculated total and partial DOS using (a) suggested U correction for Co^{2+} and Co^{3+} ions by ref. 24, (b) and (c) intermediate U correction and (d) standard DFT. (e) and (f) Calculated orbital-resolved DOS under zero U correction for Co^{2+} and Co^{3+} , respectively. (g) Calculated band structure for Co_3O_4 under zero U correction.

The electronic structure analysis indicated that, in contrast to CoO , a classic highly correlated Mott insulator, Co_3O_4 did not exhibit such strong electron correlations and its electronic structure could be well described by standard DFT. For Mott insulators the conventional DFT failed to describe the strong correlation effects between the d or f electrons and obtained a

delocalized electron distribution.^{33,34} The Hubbard U correction was thus introduced to localize the orbitals, which were then split into occupied and unoccupied states. For non-Mott insulators, the correction is often used to improve electron localization and the energetic properties. For example, Garcia-Mota *et al.*³⁵ and Wang *et al.*³⁶ found that the Hubbard U method yielded a better oxidation energy and formation enthalpy for Co_3O_4 than with plain DFT and predicted catalytic properties in good agreement with experiments. Similar cases have also been reported for other transition metal oxides.^{37–39} In some cases, the Hubbard U correction may lead to an incorrect electron distribution and a discrepancy from the actual electronic structure. One example is LaCrO_3 , for which Ong *et al.* revealed that the correction of the correlation effects for the Cr 3d states resulted in an improper electronic structure and a colorless LaCrO_3 that contradicted the experimental observation.⁴⁰ Therefore, care must be taken in applying the Hubbard U correction to transition metal oxides. Here, in Co_3O_4 , the Hubbard U correction also caused an improper electronic structure, and the experimental electronic structure and band gap of Co_3O_4 agreed well with standard DFT results.

The individual cation–anion orbital distributions near the Fermi level can be resolved from the calculations with $U_{\text{eff}} = 0$ eV. The O 2p DOS evolved into two subbands, as shown by a broad and strong band located at ~ -5 eV as well as a narrow and weak distribution near the VBM that overlapped with the metal 3d orbitals [see Fig. 3(d) and 4(d)]. A similar hybridization between the metal and ligand also occurred in the CBM. Meanwhile, the cation orbitals also exhibited complicated dispersions in both the occupied and empty states, as revealed by orbital-resolved DOS for both Co cations in Fig. 4(e) and (f). The VBM was primarily occupied by Co^{2+} e_g orbitals with

minority spin and the CBM was occupied by both Co^{2+} t_{2g} orbitals of minority spin and Co^{3+} e_g orbitals with both spin states. The further splitting of the unoccupied states was due to the coupling between the Co^{2+} and Co^{3+} 3d orbitals in the CB. The present results suggest both the orbital hybridization and crystal field effect play important roles in determining the electronic fine structure of Co_3O_4 .

Based on the obtained electronic fine structure and the energy difference between each orbital, the optical transitions in Fig. 1(a) could be directly assigned, as tabulated in Table 1. The first five transitions from peak 1 to peak 5 are pure d–d type transitions either inside a homovalent Co ion (Co^{2+} or Co^{3+}) or between two heterovalent Co ions, depending on the spin orientation. According to the dipole selection rule,⁴¹ d–d transitions would be nominally forbidden for non-hybridized d bands. However, a non-zero transition matrix element between the initial and final states may result from hybridization of the O 2p and Co 3d orbitals, which will lightly relax the parity selection rule and result in a non-zero transition probability. Nevertheless, the matrix element remained small, leading to an absorption intensity orders of magnitude lower than typically allowed transitions. The last four transitions from peak 6 to peak 9 can be attributed to ligand to metal excitations from two types of O 2p subbands to the unoccupied 3d orbitals, for both Co^{2+} and Co^{3+} in the CB. Due to the larger oscillator strength of a p–d type excitation, these transitions are orders of magnitude stronger than the d–d type transitions.⁴² We also show in Fig. 4(g) the calculated band structure of Co_3O_4 . It was found that both the VBM and CBM lie along high symmetry X points in the Brillouin zone. As a result, the minimal electronic excitation occurred along this direction without the need of momentum transfer by absorbing or emitting phonons, thus confirming the direct nature of the band gap transition observed with optical spectroscopy. However, the further specification of each band was not possible because of the heavy hybridization among the O 2p, Co^{2+} 3d and Co^{3+} 3d orbitals. The combination of the direct band gap, p-type semiconductivity and highly absorbing nature are responsible for its observed efficiency at light harvesting, electron–hole pair generation and transport upon photon excitation, which explains why Co_3O_4 is such an amazing material that exhibits excellent activity in various electro-/photo-related chemical and physical processes.

Conclusions

In summary, experimental spectroscopy and first-principles calculations were combined to reveal the electronic structure of Co_3O_4 , which clarified two decades of discrepancies regarding the nature of the band gap and the characteristic optical transitions. We demonstrated evidence of a direct fundamental band gap of 0.76 eV in the NIR that is significantly smaller than previously reported values. The observed electronic excitations were found to originate from the coupling of the crystal field splitting between the tetrahedral and octahedral site cations and from a cation–anion orbital hybridization. It was shown that Co_3O_4 did not exhibit a strong electron correlation as compared with the classic CoO and the electronic structure and semiconducting nature of Co_3O_4 obtained from standard DFT agreed well with the experimental results. The resolved electronic structure of Co_3O_4 illustrated the reason that this material exhibits excellent optoelectronic, photocatalytic, and artificial photosynthesis properties and may assist in the design of nano- and hetero-structured devices based on this material. Finally, the method we demonstrated to resolve the electronic and optical properties of this materials can be extended to the functional properties of other complex transition metal oxides and may help to identify and resolve cases where the electronic structure has been oversimplified.

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Table 1 Optical transition assignments for epitaxial Co_3O_4 as determined in Fig. 1(a)

Peak	Energy (eV)	Transitions	Intensity (arb. units)	Type
1	0.83	$\text{Co(II)}\ 3d\ e_g \downarrow \rightarrow \text{Co(II)}\ 3d\ t_{2g} \downarrow$	1	d–d (direct)
2	0.95	$\text{Co(II)}\ 3d\ e_g \downarrow \rightarrow \text{Co(III)}\ 3d\ e_g \downarrow$	0.73	d–d
3	1.04	$\text{Co(II)}\ 3d\ t_{2g} \uparrow \rightarrow \text{Co(III)}\ 3d\ e_g \uparrow$	0.33	d–d
4	1.64	$\text{Co(III)}\ 3d\ t_{2g} \downarrow \rightarrow \text{Co(II)}\ 3d\ t_{2g} \downarrow$	6.86	d–d
5	1.84	$\text{Co(III)}\ 3d\ t_{2g} \rightarrow \text{Co(III)}\ 3d\ e_g$	6.92	d–d
6	2.62	$\text{O}\ 2p(\text{I}) \downarrow \rightarrow \text{Co(II)}\ 3d\ t_{2g} \downarrow$	63.8	p–d
7	3.66	$\text{O}\ 2p(\text{I}) \rightarrow \text{Co(III)}\ 3d\ e_g$	73.3	p–d
8	4.69	$\text{O}\ 2p(\text{II}) \downarrow \rightarrow \text{Co(II)}\ 3d\ t_{2g} \downarrow$	51.7	p–d
9	5.8	$\text{O}\ 2p(\text{II}) \rightarrow \text{Co(III)}\ 3d\ e_g$	192	p–d

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